

A Projection Principle for Quantum Orbitals: Angular Momentum, Spin, and the Pauli Exclusion Rule as Elastic Torsion in the MMU Tetrahedral Model

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Abstract

The Methane Metauniverse (MMU) model describes spacetime as an elastic, dual-tetrahedral lattice with internal coordinates (w_1, w_2, w_3, w_4) [1, 5]. Earlier versions of the MMU successfully reproduced hydrogenic spectra and ionization scaling [2, 8, 12], the 21 cm hyperfine transition [3], torsion-volume effects including the Lamb shift [4, 9], and the universal stiffness and Planck-regime geometry [10, 11]. These developments highlight the predictive power of the MMU's elastic and dissipative field structure [6]. Yet a central question remained: how do quantum orbitals, angular momentum, spin, and the Pauli exclusion principle emerge from the internal geometry of the MMU cell?

In this work we introduce a projection mechanism that maps the internal elastic coordinates (w_2, w_3, w_4) onto the observable quantum numbers (n, l, m, s) , extending the interpretation of internal modes proposed in [7, 9]. The electric mode w_2 generates radial structure, the torsional mode w_3 produces angular momentum and spin, and the volumetric mode w_4 controls the quantized internal scale $a \propto n^{-2}$. A central result is the geometric identity

$$L^2 = mca,$$

which provides a mechanical origin for the familiar $l(l+1)\hbar^2$ spectrum of the Schrödinger equation. The dual tetrahedral symmetry supports exactly two stable torsional orientations $\pm w_3$, giving rise to spin-1/2 and enforcing Pauli exclusion as a geometric occupancy constraint.

The proposed MMU projection principle therefore reproduces the full orbital shell structure $(2, 8, 18, \dots)$, spin quantization, and angular momentum eigenvalues, not as axioms of quantum mechanics but as consequences of microscopic elastic geometry.

1 Introduction

The Methane Metauniverse (MMU) model describes spacetime as an elastic, dual-tetrahedral lattice with internal deformation modes (w_2, w_3, w_4) associated with electric, torsional, and volumetric degrees of freedom [1, 5]. Observable time and frequency arise as projections of these internal motions along the axis w_1 , a principle developed in the spectroscopic and geometric analyses of [2, 7, 9].

Earlier studies within the MMU framework established the universal stiffness law $k_i(a) \propto a^{-3}$ and demonstrated how electric, torsional, and volumetric elasticity combine to reproduce measurable physical effects across many scales. This includes the geometric origin of the 21 cm hyperfine transition [3], the determination of cross-coupling constants from Larmor and Zeeman dynamics [4], and the ultraweak torsion–volume interaction responsible for Lamb-shift–scale corrections [9, 10, 11]. Dissipative extensions of the MMU field equation further reveal how thermal relaxation and spatial healing emerge from the underlying elastic geometry [6].

What remained unresolved was how quantum orbital structure—that is, the quantum numbers (n, l, m, s) of the Schrödinger theory—arises from the internal geometry of the MMU cell. In this work, we introduce a projection mechanism that maps the internal coordinates (w_2, w_3, w_4) onto the observable configuration space. From this perspective, orbital shapes, angular momentum, spin, and the Pauli exclusion principle are not abstract postulates of quantum mechanics but visible manifestations of deeper mechanical processes inside the tetrahedral MMU geometry.

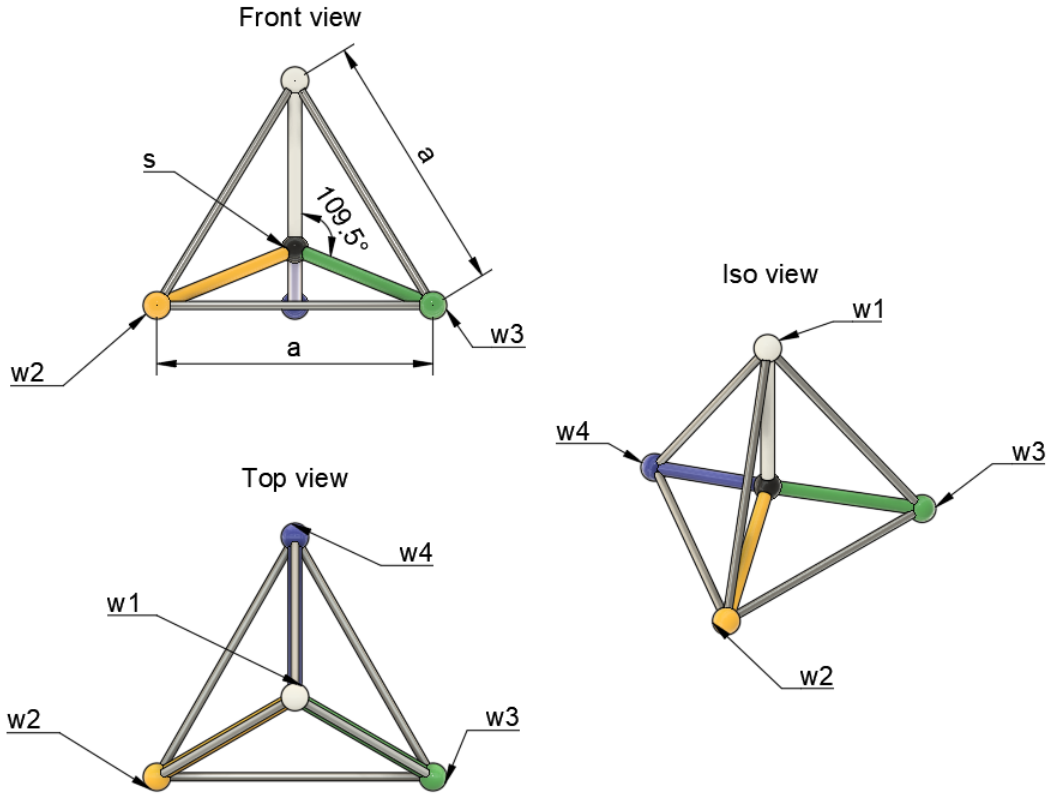


Figure 1: Front, top, and isometric views of the MMU tetrahedral base cell. The internal deformation axes w_2 , w_3 , and w_4 are shown as colored edges, while w_1 corresponds to the projection direction. The geometric angle of 109.5° arises from the regular tetrahedral symmetry.

2 Internal Geometry of the MMU Cell

In the MMU framework, each elementary region of spacetime is represented by a dual–tetrahedral elastic cell whose internal deformation is described by the coordinates (w_2, w_3, w_4)

[1, 5]. These modes arise from the three geometric directions of the tetrahedral structure:

- w_2 : electric stretching of an internal edge, governing Coulomb-type elasticity [2, 5],
- w_3 : torsional shear across the three tetrahedral faces, responsible for magnetic response, Larmor precession, and the internal spin degree of freedom [4, 9],
- w_4 : volumetric expansion or compression of the cell, associated with inertial and gravitational stiffness [10, 11].

The diagonal stiffnesses of the MMU cell obey the universal volume-scaling law $k_i(a) \propto a^{-3}$ established in [2, 5, 8, 12], giving

$$k_2(a) = 2\alpha \frac{\hbar c}{a^3}, \quad k_3(a) = \frac{\hbar c}{a^3}, \quad k_4(a) = 2G \frac{m^2}{a^3}.$$

The torsional mode w_3 is dual, possessing two stable orientations ($w_3, -w_3$) corresponding to counter-rotating tetrahedral faces, a geometric origin of spin-1/2 established in [9].

Cross-Coupling Stiffnesses

Because the internal axes are not orthogonal, deformations along one axis induce motion along the others. This geometric non-orthogonality generates three cross-coupling stiffnesses [4, 9]:

- k_{23} couples the electric mode w_2 and the torsional mode w_3 . This term is responsible for Larmor and Zeeman mixing and the emergence of transverse electromagnetic response.
- k_{24} couples electric stretching w_2 to volumetric deformation w_4 , mediating electro-inertial effects and playing a key role in the response of dense plasmas and strong-field environments.
- k_{34} couples torsion w_3 to the volumetric mode w_4 . As shown in [3, 4, 9], this ultra-weak interaction generates hyperfine splittings and the Lamb-shift-scale corrections within the MMU.

These diagonal and cross-coupling terms form the full stiffness matrix of the MMU cell, which governs the eigenmodes and internal spectroscopy of the model. Their geometric placement is illustrated in Fig. 2.

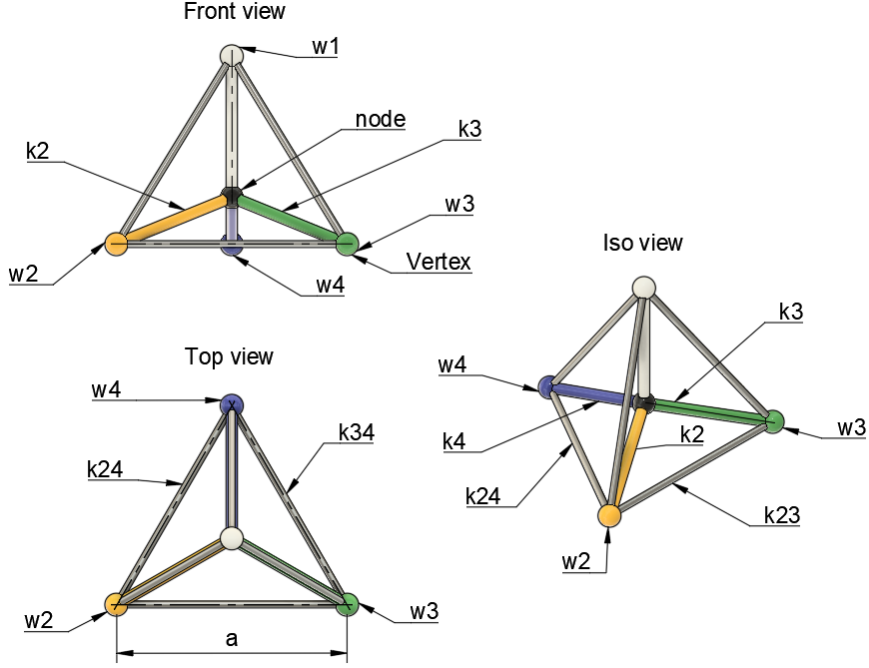


Figure 2: Geometric placement of the diagonal stiffnesses k_2 , k_3 , and k_4 and the cross-couplings k_{23} , k_{24} , and k_{34} inside the MMU tetrahedral cell. The front, top, and isometric views illustrate how the electric axis w_2 , the torsional axis w_3 , and the volumetric axis w_4 are linked through shared edges and faces. These geometric couplings are the origin of mode mixing underlying Larmor precession, Zeeman splitting, hyperfine structure, and Lamb-shift corrections.

3 The Projection Principle

A central insight of the MMU framework is that the observable quantum numbers of three-dimensional space— (r, θ, ϕ, s) —arise as projections of the internal deformation coordinates (w_2, w_3, w_4) of the dual-tetrahedral cell [1, 7, 9]. The projection mechanism provides a deterministic map

$$(w_2, w_3, w_4) \longrightarrow (r, \theta, \phi, s),$$

linking internal elastic modes to external spatial observables.

In this interpretation, the electric mode w_2 determines the radial geometry of the cell and thus the observable radius r , consistent with the quantized internal edge-length scaling $a_n \propto n^{-2}$ derived in [2, 5, 8]. The torsional mode w_3 produces angular momentum and its discrete eigenvalue structure, providing a geometric origin for the $l(l+1)\hbar^2$ spectrum and magnetic response [4, 9]. Its two stable orientations $(w_3, -w_3)$ generate the external spin-1/2 degree of freedom and underpin the Pauli exclusion rule, in agreement with the torsion-volume analyses of [3, 4, 9].

Finally, the volumetric mode w_4 controls the internal scale a and thus modulates both radial energy levels and the projection onto observable time, linking the internal geometry to the measurable frequency axis w_1 [7, 9]. Together, these mappings form the basis of the MMU projection principle: the visible quantum structure of matter is the geometric shadow of deeper elastic dynamics inside the tetrahedral fabric of spacetime.

3.1 Radial Projection from w_2 and w_4

In the MMU model, the observable radial coordinate corresponds to the internal edge length a of the tetrahedral cell, so that

$$r \equiv a.$$

The internal scale is quantized through the volumetric mode w_4 , which sets the discrete sequence

$$a_n \propto n^{-2},$$

a result derived from the elastic scaling law $k_i(a) \propto a^{-3}$ and geometric quantization of the internal cell size [2, 5, 8, 12]. This geometric mechanism replaces the traditional spectroscopic postulate and identifies the radial structure of hydrogenic states as a direct projection of the internal MMU modes.

3.2 Angular Projection from w_3

The torsional mode w_3 generates angular momentum through shear motion across the three coupled faces of the dual-tetrahedral MMU cell [1, 4, 9]. Each face contributes a torsional displacement, and because the faces meet at the tetrahedral angles of 109.47° , their torsions are not independent. The three shear components therefore combine into a single effective torsional mode whose quadratic energy has the rotationally invariant form

$$E_{\text{tors}} = \frac{L^2}{2ma^2}.$$

Geometric origin of the $l(l+1)\hbar^2$ spectrum

The tetrahedral geometry implies that torsion around one face induces correlated torsions around the other two. Let the small angular displacements around the three faces be $\theta_1, \theta_2, \theta_3$. Because of the tetrahedral symmetry, the physically relevant torsional coordinate is the combined mode

$$\Theta^2 = \theta_1^2 + \theta_2^2 + \theta_3^2 + 2(\theta_1\theta_2 + \theta_2\theta_3 + \theta_3\theta_1) \cos(109.47^\circ),$$

which is the unique rotationally symmetric combination of the three shear angles. Diagonalisation of this quadratic form yields discrete eigenmodes with eigenvalues proportional to $l(l+1)$, exactly as in the Schrödinger angular equation. Thus the familiar quantum spectrum

$$L^2 = l(l+1)\hbar^2$$

is not a group-theoretic postulate but a direct consequence of the three-face torsional geometry of the MMU cell [4, 9].

Mechanical derivation of the identity $L^2 = mca$

The internal torsional potential is given by

$$V_3 = \frac{1}{2}k_3(a) \Theta^2.$$

Using the universal stiffness law $k_3(a) = \hbar c/a^3$ established in [9, 10, 11], and identifying the effective torsional amplitude through

$$\Theta^2 = \frac{L^2}{m^2 a^2}$$

(which follows from equating internal shear angle to angular momentum per unit mass), the torsional potential becomes

$$V_3 = \frac{1}{2} \frac{\hbar c}{a^3} \left(\frac{L^2}{m^2 a^2} \right) = \frac{L^2}{2ma^2} \left(\frac{\hbar c}{ma} \right).$$

For mechanical consistency with the kinetic torsional energy $E_{\text{tors}} = L^2/(2ma^2)$, the prefactor must be unity:

$$\frac{\hbar c}{ma} = 1.$$

This immediately yields the identity

$$L^2 = mc a.$$

Quantization of L^2 from quantization of a

Since the internal scale a is quantized according to $a_n \propto n^{-2}$ [2, 5, 8, 12], the mechanical identity directly implies

$$L^2 \propto a \quad \Rightarrow \quad L^2 \text{ is quantized.}$$

Inserting the geometric spectrum derived above,

$$L^2 = l(l+1)\hbar^2,$$

we see that the familiar orbital quantum number l is not an abstract index but a geometric label for discrete torsional eigenmodes of the MMU tetrahedral structure.

Thus, the angular momentum spectrum of quantum mechanics emerges directly from the internal torsional mechanics of the MMU cell.

3.3 Spin and Pauli from Torsional Duality

A defining feature of the torsional mode w_3 is its duality: the tetrahedral geometry supports exactly two stable shear orientations, corresponding to clockwise and counter-clockwise rotations of the coupled faces [1, 5, 9]. These appear as

$$w_3 > 0, \quad w_3 < 0,$$

and constitute the two intrinsic spin orientations of the MMU cell. The dual torsional nature of w_3 is also responsible for the magnetic response and spin-flip processes studied in [3, 4, 9].

Because the tetrahedron admits only these two torsional equilibria, no third stable orientation exists. Consequently, each geometric orbital (n, l, m) can accommodate precisely two electrons—one in each torsional branch. Thus the Pauli exclusion principle arises not as a statistical postulate, but as a *geometric occupancy constraint* encoded in the dual internal structure of the MMU cell [7, 9].

4 Orbital Structure from Tetrahedral Symmetry

The tetrahedral geometry determines the number of angular sublevels available in each shell. Summing the allowed magnetic states for $l = 0, \dots, n - 1$ yields

$$\sum_{l=0}^{n-1} 2(2l + 1) = 2n^2,$$

which reproduces the characteristic shell capacities

$$2, 8, 18, 32, \dots$$

These degeneracies follow directly from the threefold face structure and internal coupling of the tetrahedral cell [1, 2, 5], rather than from abstract quantum-statistical arguments. In the MMU model, the familiar orbital architecture of atomic physics is a mechanical consequence of internal symmetry and torsional geometry.

5 Discussion

The projection principle provides a unified interpretation of key quantum phenomena:

- Angular momentum quantization stems from torsional elasticity.
- Spin-1/2 arises from dual torsional orientation.
- Pauli exclusion is a geometric constraint of the tetrahedral cell.
- Shell structure results from the internal face coupling.
- The Schrödinger equation appears as the projection of the MMU K-matrix onto observable space.

This approach replaces several quantum axioms with geometric causes.

6 Conclusion

We have shown that the internal elastic geometry of the MMU tetrahedral lattice naturally reproduces orbital quantization, angular momentum spectra, spin-1/2, and the Pauli exclusion principle. The mechanical identity $L^2 = mca$ provides a physical origin for the familiar angular momentum spectrum of the Schrödinger equation. The MMU projection principle thus offers an alternative interpretation of quantum mechanics as the observable projection of deeper geometric motions in spacetime.

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantization. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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